
Computational Lithography Solutions for Non-Immersion ArF (193nm) Excimer Laser Deep Ultraviolet Machines for Chemically Amplified Resists with Bottom Anti-Reflective Coating

Executive Program in Semiconductor Manufacturing -
IIT Delhi

Tejas Beedkar
Sushma Shetty
Nitesh Mahawar
Juganta Mosrong

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6.1. Role of the Projection Lens in Reduction Lithography

In projection photolithography, the projection lens is the core imaging element that transfers the pattern from the mask (reticle) to the wafer with a fixed reduction ratio, typically 4:1 or 5:1. It replaces earlier contact and proximity methods, where the mask sat close to the wafer and suffered from mask damage, particle-induced defects, and limited resolution. By separating mask and wafer and using an imaging lens, projection systems achieve smaller features, better overlay,

and greatly improved mask lifetime compared with contact printers. Reduction also relaxes mask requirements: for a 4× system, a 400 nm feature on the mask prints as 100 nm on the wafer, making mask fabrication and inspection more feasible while reducing the effective size of mask defects on wafer.

The projection lens not only reduces the image but also defines key optical parameters. It sets the numerical aperture (NA), which, together with wavelength, determines resolution through relations of the form

Lithographic resolution is approximately:

$$Resolution = k_1 \frac{\lambda}{NA}$$

where: λ = exposure wavelength NA = numerical aperture k_1 = process factor The projection lens largely determines the achievable NA.

and depth of focus (DOF) through

$$DOF = k_2 \frac{\lambda}{NA^2}$$

Higher NA improves resolution but reduces focus margin and enables finer resolution at the cost of reduced DOF. The lens is designed to minimize aberrations such as spherical aberration, coma, astigmatism, field curvature, distortion, and chromatic effects; even nanometer-level distortion can impact device performance, so multi-element high-NA designs are used to keep the aerial image close to ideal.

6.2. Computational Lithography and Mask Optimization

Modern imaging performance does not rely on the projection lens alone but on a full computational pipeline that models the lens and uses that model to design the mask. In optical proximity correction (OPC) and broader computational lithography, the mask layout is modified—using serifs, biasing, sub-resolution assist features (SRAFs), and phase-shifting mask (PSM) techniques—so that, after propagation through the known projection lens, the printed wafer pattern matches the target critical dimensions (CDs) and shapes. The lens is characterized as a transfer function (including NA, aberrations, and illumination coupling), and mask synthesis uses this transfer function to pre-compensate for imaging imperfections.

Because the projection lens is fixed in hardware and extremely expensive to modify, computational lithography becomes the primary means of “virtual” tuning. Different products and layers can use different OPC recipes, assist feature placements, or PSM configurations tailored to specific pitches and orientations, all computed with the same lens model. This allows a single lens to support a wide variety of patterns while maintaining tight CD control and process window, particularly at advanced nodes where purely optical tuning is no longer sufficient.

[1]

6.3. Resist Process Co-Optimization and Depth of Focus

Even with an optimized lens and OPC, the resist process strongly influences usable DOF and line-edge quality. Co-optimizing resist thickness, exposure, post-exposure bake (PEB), and development conditions can widen the effective process window and mitigate the impact of lens-limited DOF. For example, adjusting resist thickness or index can reduce standing waves and improve focus latitude, while modified PEB conditions can tune acid diffusion and image contrast.

Developer chemistry and process conditions (time, temperature, agitation) further shape the final profile, influencing CD, sidewall angle, and line edge roughness. By aligning resist stack design with the optical parameters of the lens—particularly NA and focus behavior—engineers can trade off sensitivity, DOF, and CD uniformity in a controlled way. In practice, resist and lens are treated as a combined imaging system: once the lens design is fixed, resist process development becomes a major lever to recover DOF and robustness without changing the hardware.

[2]

6.4. Illumination and Recipe Selection Around the Lens

Although the projection lens is central, its performance depends strongly on the illumination system and scanner operating mode. Different source shapes—conventional, annular, quadrupole, dipole, or custom free-form sources—fill the lens pupil differently and therefore change contrast and process window for specific pitches and orientations. By choosing appropriate combinations of NA and partial coherence (σ), engineers can emphasize certain diffraction orders and suppress others, improving imaging for dense or isolated features as needed.

Recipe selection also includes scanner mode choices such as stepper versus step-and-scan, slit orientation, and field layout. These choices affect how field-dependent lens aberrations and slit-scan dynamics project onto the wafer. For a given lens, one layer may favor higher NA with a tailored off-axis illumination, while another layer uses lower NA and more conventional illumination to gain DOF. Thus, illumination and recipe parameters act as “soft” knobs that shape how the fixed projection lens is used for each product and layer.

[3]

6.5. Practical Fine-Tuning of Projection Lenses in Fabs

In real manufacturing, “fine tuning the projection lens” rarely means physically redesigning or replacing optics. Instead, it involves keeping the lens close to its calibrated ideal and compensating any residual, stable signatures through scanner and process control. High-mileage

lenses can suffer from thermal effects: high-energy exposure heats optical elements, inducing focus drift, aberration changes, and image degradation. Mitigation strategies include lens-heating control, optimized illumination conditions, and active correction using lens manipulators or wavefront control systems.

Scanner knobs—such as focus and leveling control, magnification and scaling adjustments, and sometimes low-order aberration actuators—are used to maintain overlay and CD performance as the lens ages or operating conditions change. At the same time, OPC/computational lithography and resist/illumination optimization are updated to account for measured, stable lens signatures instead of forcing the lens to be mathematically perfect. In this combined scheme, hardware tuning keeps the lens within a defined optical spec, while process and computational tuning tailor imaging for each layer, achieving the required resolution, overlay, and yield without constant physical modification of the lens assembly.

[\[4\]](#)

6.6 Primary Failure Modes with Respect to the Projection Lens

For modern DUV projection lenses, the dominant failure mechanisms are usually:

6.6.1 Fundamental Resolution / NA Limit

The projection lens can only collect diffraction orders up to its NA; once required features demand spatial frequencies outside that NA, image contrast collapses and no amount of process tweaking can fully recover CDs. This is the classic Rayleigh limit: as you push to smaller pitches at fixed wavelength and NA, the lens becomes the bottleneck because higher diffraction orders never reach the wafer.

[\[5\]](#)

6.6.2 Thermal Distortion and Drift of the Lens

Repeated high-energy pulses heat lens elements by millikelvins and move them by nanometers, which is already enough to change focus, astigmatism, and field distortion and to cause CD and overlay errors. If active actuators and calibration cannot fully compensate (for example on “high-mileage” tools), the projection lens becomes a primary reliability limiter for CD uniformity and overlay over long lots.

[\[4\]](#)

6.6.3 Residual and Evolving Aberrations

Even after factory optimization, a real lens has residual low- and mid-order aberrations; with time, contamination, coating changes, or mechanical creep can increase these. When aberration terms exceed what can be compensated by scanner manipulators and OPC, they show up as systematic

CD signatures, across-field focus variation, and overlay distortion that fundamentally trace back to the lens design or health.

[\[6\]](#)

In short, the projection lens “fails” primarily when its NA and aberration characteristics no longer allow enough contrast and process window for the target pitch and overlay, either because physics is exhausted (NA limit) or because heating/aging push the real lens away from its designed performance. Everything done with OPC, illumination, and resist is essentially an attempt to postpone or work around those lens-centric failure points.

[\[5\]](#)

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1. **Projection photolithography (book chapter)**

Classical Optics and its Applications, “Projection photolithography” (Cambridge University Press).

2. **Aspherical lens design using computational lithography**

Journal article on aspherical lens design and NA setting for lithography (ScienceDirect).

3. **Projection photolithography (slide deck)**

“Projection photolithography” presentation hosted on SlideShare.

4. **Reducing lens heating effects on high mileage projection lenses used in optical lithography**

Paper from CS MANTECH on projection-lens heating and image degradation.

5. **Photolithography – general overview**

Photolithography article from an online encyclopedia covering the general process, including contact vs. projection lithography.

6. **Microscope / projection photolithography papers**

Academic papers on microscope-based projection photolithography and rapid prototyping, including UV LED-based setups.

7. **Lenses & mirrors – lithography principles**

ASML technical page describing lithography lenses, mirrors, and NA/resolution relationships.

8. **Understanding focus effects and DOF in optical lithography**

Technical paper on focus, log-slope, and depth of focus behavior in submicron optical lithography.

9. **Photolithography limitations and process window discussions**

Educational material on photolithography limits, including resolution, DOF, and process window.

10. Comprehensive model for projection photolithography

Thesis or technical report modeling the scanner + resist system, including lens errors and systematic effects.

11. Exhausting the resolution limits of 193-nm projection lithography

Journal article on pushing 193-nm lithography with computational lithography and lens limits.

12. Optical lithography – historical perspective

Review paper on the evolution of optical lithography and projection lenses.

13. Lithography optics basics

Lecture or tutorial PDF on lithography optics, including NA, aberrations, and projection lens behavior.

14. Intro to photolithography (industry training PDF)

Industry training material, such as Micron documentation, describing projection optics and 4× reduction.